Angles of elevation and depression

Core Concepts: Angles of elevation and depression

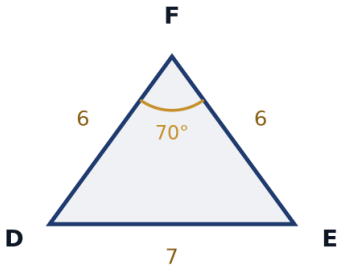
1 In $\triangle ABC$ studying angles of elevation and depression, $m\angle A = 52^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 83°

b. 93°

c. 75°

d. 128°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$?

a. 8

b. 15

c. 13

d. 18

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying angles of elevation and depression, $m\angle A = 52^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 76°

b. 86°

c. 68°

d. 128°

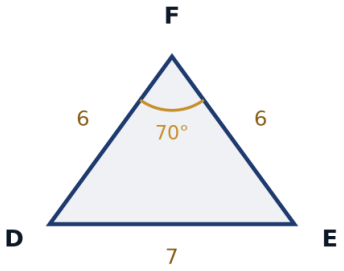
5 A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm

c. 24 cm

d. 54 cm



6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 13

b. 20

c. 18

d. 23

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying angles of elevation and depression, $m\angle A = 52^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 69°

b. 79°

c. 61°

d. 128°

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8 A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm

c. 30 cm

d. 96 cm

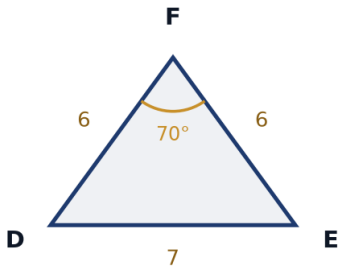
9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$?

a. 18

b. 25

c. 23

d. 28



Exam-Style Problem Solving & Modeling

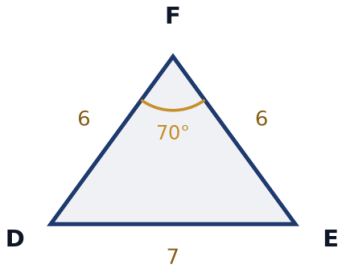
10 In $\triangle ABC$ studying angles of elevation and depression, $m\angle A = 52^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 62°

b. 72°

c. 54°

d. 128°



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11 A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm

c. 36 cm

d. 150 cm

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 23

b. 30

c. 28

d. 33

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